

R for bioinformatics, data visualisation, part 2

HUST Bioinformatics course series

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Section 1

section 1: TOC

前情提要

vector & matrix:

- declaration
- manipulation
- arithmetic
- transposition

vectorization

- every is a vector!!
- vectorization verses loop (will be explained later)
- advantages using vectorization

今次预报

- ① data.frame, tibble
- ② read files from harddrive (IO)
- ③ exercises & homework

Section 2

section 2: data.frame and tibble

data.frame, outline

- ① what is a data.frame???
- ② how to make a data.frame
- ③ how to add row(s)/col(s) to an existing data.frame how to combine two data.frames
- ④ how to manipulate a data.frame

2.1 what is a data.frame?

眼见为实:

```
library(tidyverse); ## 装入包
library(kableExtra);
kbl( head(mpg), booktabs = T); ## 显示前几行数据
```

manufacturer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class
audi	a4	1.8	1999	4	auto(l5)	f	18	29	p	compact
audi	a4	1.8	1999	4	manual(m5)	f	21	29	p	compact
audi	a4	2.0	2008	4	manual(m6)	f	20	31	p	compact
audi	a4	2.0	2008	4	auto(av)	f	21	30	p	compact
audi	a4	2.8	1999	6	auto(l5)	f	16	26	p	compact
audi	a4	2.8	1999	6	manual(m5)	f	18	26	p	compact

注意 head() tail() 的用法和参数



head 和 tail 的用法

```
nrow(mpg); ## total number of rows
```

```
[1] 234
```

```
kbl( head(mpg, n=3), booktabs = T); ## 显示前几行数据
```

manufacturer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class
audi	a4	1.8	1999	4	auto(l5)	f	18	29	p	compact
audi	a4	1.8	1999	4	manual(m5)	f	21	29	p	compact
audi	a4	2.0	2008	4	manual(m6)	f	20	31	p	compact

```
kbl( tail(mpg, n=3), booktabs = T); ## 显示最后 3 行数据
```

manufacturer	model	displ	year	cyl	trans	drv	cty	hwy	fl	class
volkswagen	passat	2.8	1999	6	auto(l5)	f	16	26	p	midsize
volkswagen	passat	2.8	1999	6	manual(m5)	f	18	26	p	midsize
volkswagen	passat	3.6	2008	6	auto(s6)	f	17	26	p	midsize

data.frame 的组成与常用函数

组成

- 二维表格
- 由不同列组成；每列是一个 **vector**，不同列的数据类型可以不同，但一列只包括一种数据类型（int, num, chr ...）
- 各列的长度相同

常用 functions

- `nrow( )`;
- `ncol( )`;
- `dim( )`;
- ...

structure of data.frame: str 函数

```
str( mpg );
```

```
tibble [234 x 11] (S3: tbl_df/tbl/data.frame)
 $ manufacturer: chr [1:234] "audi" "audi" "audi" "audi" ...
 $ model       : chr [1:234] "a4" "a4" "a4" "a4" ...
 $ displ      : num [1:234] 1.8 1.8 2 2 2.8 2.8 3.1 1.8 1.8 2 ...
 $ year       : int [1:234] 1999 1999 2008 2008 1999 1999 2008 1999 1999 2008 ...
 $ cyl        : int [1:234] 4 4 4 4 6 6 6 4 4 4 ...
 $ trans      : chr [1:234] "auto(l5)" "manual(m5)" "manual(m6)" "auto(av)" ...
 $ drv        : chr [1:234] "f" "f" "f" "f" ...
 $ cty        : int [1:234] 18 21 20 21 16 18 18 18 16 20 ...
 $ hwy        : int [1:234] 29 29 31 30 26 26 27 26 25 28 ...
 $ fl         : chr [1:234] "p" "p" "p" "p" ...
 $ class      : chr [1:234] "compact" "compact" "compact" "compact" ...
```

注: Tibble class 是 data.frame 的升级版本; 本课程将二者混用, 以 tibble 为主。用?mpg 命令查看 mpg 各列的意义



2.2 make a new data.frame

使用 data.frame 函数创建新的 data.frame:

```
## data.frame()
( dat2 <-
  data.frame( "data.d" = sample( 1:100, 10 ),
             group = sample( LETTERS[1:3], 10, replace = TRUE),
             data2 = 0.1 )
);
```

	data.d	group	data2
1	49	A	0.1
2	9	B	0.1
3	90	B	0.1
4	88	A	0.1
5	12	B	0.1
6	71	B	0.1
7	33	A	0.1
8	57	C	0.1
9	46	C	0.1
10	41	C	0.1

2.2 make a new data.frame, cont.

```
str(dat2);
```

```
'data.frame':  10 obs. of  3 variables:
 $ data.d: int  49 9 90 88 12 71 33 57 46 41
 $ group : chr  "A" "B" "B" "A" ...
 $ data2 : num  0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1
```

2.3 how to add row(s)/col(s) to an existing data.frame

先创建”表头”，再填充

```
df2 <- data.frame( x = character(), y = integer(), z = double() , stringsAsFactors = FALSE );

##
df2 <- rbind( df2, data.frame( x = "a", y = 1L, z = 2.2 ) );
df2 <- rbind( df2, data.frame( x = "b", y = 2, z = 4.4 ) );

df2;
```

```
  x y    z
1 a 1 2.2
2 b 2 4.4
```

注意

- 使用 `rbind` 函数
- 新的一行用 `data.frame` 定义，其”表头”需要与合并表相同

问题:

以下代码能顺利执行吗?

```
## 注意这里的 data.frame 会有多行 ...  
df2 <- rbind( df2, data.frame( x = c("a","b","c"), y = 1L, z = 2.2 ) );
```

data.frame, add column

用 cbind 函数增加列: **column bind**

```
m <- cbind(1, 1:7) ; ## 产生两列数据 7 行数据 ..
( m <- cbind(m, 8:14) ); ## 增加一列 也有 7 行数据 ...
```

	[,1]	[,2]	[,3]
[1,]	1	1	8
[2,]	1	2	9
[3,]	1	3	10
[4,]	1	4	11
[5,]	1	5	12
[6,]	1	6	13
[7,]	1	7	14

data.frame, add column, cont.

自行练习，回答代码中的问题:

```
## 1. 生成一个 10 行 2 列的 data.frame  
df3 <- data.frame( data = 1:10, group = c("A","B") );
```

```
## 2. 增加一列，其长度是 1，可以吗？  
cbind(df3, newcol = 1);
```

```
## 3. 增加一列，其长度是 10，可以吗？  
cbind(df3, newcol = 1:10);
```

```
## 4. 增加一列，其长度是 2，可以吗？  
cbind(df3, newcol = 1:2);
```

```
## 5. 增加一列，其长度是 3，可以吗？  
cbind(df3, newcol = 1:3);
```

data.frame, 以列方式合并两个 data.frame

同样使用 cbind

```
df4 <- data.frame( data = 1:10, group = c("A","B") );
df5 <- data.frame( length = sample(1:100, 10), width = sample(1:100, 10) );

## --
head( cbind( df4, df5 ) );
```

	data	group	length	width
1	1	A	74	4
2	2	B	62	62
3	3	A	96	80
4	4	B	90	41
5	5	A	29	58
6	6	B	11	13

如果 一个 df 的行数少于另一处怎么办?

```
df6 <- data.frame( length = sample(1:100, 5), width = sample(1:100, 5) );
head( cbind( df4, df6 ) );
```

	data	group	length	width
1	1	A	41	18



2.4 how to manipulate a data.frame

自行尝试以下操作

```
## 取行:
df4[ 1:2, ];

## 取列
df4[, 2]

## 取行, 重新排序
df4[c(2,3,1), ]

## 取列, 重新排序
df4[, c(2,1)]

## 替换一行:
df4[1, ] <- data.frame( data = 100, group = "A" );

## 替换一列:
df4[, "data"] <- sample( 1:100, 5 );
```

tibble, outline

- 5 how to make a tibble
- 6 how to add row(s)/col(s) to an existing tibble how to combine two tibble
- 7 how to manipulate a tibble

2.5 make a new tibble

tibble 相关功能由 tibble 或 tidiverse 包提供

```
library(tibble); ## 或 library(tidiverse);
## 用 tibble 函数创建, 用法和 data.frame() 相似
( dat <-
  tibble( data = sample( 1:100, 10 ),
          group = sample( LETTERS[1:3], 10, replace = TRUE),
          data2 = 0.1 )
);
```

```
# A tibble: 10 x 3
  data group data2
  <int> <chr> <dbl>
1     71 C      0.1
2      6 B      0.1
3     25 A      0.1
4     63 B      0.1
5     35 A      0.1
6     33 C      0.1
7      5 B      0.1
8     32 C      0.1
9     88 A      0.1
10    100 A      0.1
```

str(dat)

查看得到的数据结构

```
str(dat);
```

```
tibble [10 x 3] (S3: tbl_df/tbl/data.frame)
 $ data : int [1:10] 71 6 25 63 35 33 5 32 88 69
 $ group: chr [1:10] "C" "B" "A" "B" ...
 $ data2: num [1:10] 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1
```

创建 tibble 的另一种方式 (by row)

```
## 另一种方式创建 tibble
tribble(
  ~x, ~y, ~z,
  "a", 2, 3.6,
  "b", 1, 8.5
)
```

```
# A tibble: 2 x 3
  x       y       z
<chr> <dbl> <dbl>
1 a         2   3.6
2 b         1   8.5
```

2.6 how to add row(s)/col(s) to an existing tibble

```
## 新 tibble, with defined columns ... 创建表头
tb <- tibble( x = character(), y = integer(), z = double() );
dim(tb);
```

```
[1] 0 3
```

```
## 增加行 ...
tb <- add_row( tb, x = "a", y = 2, z = 3.6 );
tb <- add_row( tb, x = "b", y = 1, z = 8.5 );
```

```
## 显示
tb;
```

```
# A tibble: 2 x 3
  x         y     z
<chr> <dbl> <dbl>
1 a         2   3.6
2 b         1   8.5
```



tibble, add_row 插入时指定行号

```
## 生成一个 tibble
df <- tibble(x = 1:3, y = 3:1);

# 在第二行之前插入
df <- add_row(df, x = 4, y = 0, .before = 2);

df;
```

```
# A tibble: 4 x 2
```

	x	y
	<dbl>	<dbl>
1	1	3
2	4	0
3	2	2
4	3	1

tibble, add_row 插入多行, 插入另一个 tibble??

```
## 插入多行
df <- add_row(df, x = 4:5, y = 0:-1);

## 插入另一个 tibble (与另一个 tibble 合并) ???
df2 <- tibble( x = as.double(200:202), y = as.double(1000:1002) );
df3 <- add_row( df, df2 ); ## 可以运行 ...
```

tibble, 合并多个 tibble

```
df3 <- bind_rows( df, df2 );  
df3;
```

```
# A tibble: 9 x 2
```

	x	y
	<dbl>	<dbl>
1	1	3
2	4	0
3	2	2
4	3	1
5	4	0
6	5	-1
7	200	1000
8	201	1001
9	202	1002

tibble, add column

```
tb3 <- tribble(
  ~x, ~y, ~z,
  "a", 2, 3.6,
  "b", 1, 8.5
);

tb3 <- add_column( tb3, a = 98 ); ## recycle ...
tb3 <- add_column( tb3, b = LETTERS[1:2], c = c("CHEN", "WANG") );
tb3;
```

```
# A tibble: 2 x 6
```

	x	y	z	a	b	c
	<chr>	<dbl>	<dbl>	<dbl>	<chr>	<chr>
1	a	2	3.6	98	A	CHEN
2	b	1	8.5	98	B	WANG

tibble, 按列合并两个 tibble?

练习:

- 1 尝试用 `add_column` 合并两个 tibble
- 2 使用 `bind_cols` 合并两个 tibble

2.7 how to manipulate a tibble

自行练习以熟悉以下操作:

```
## 取得行
```

```
tb3[c(1,2), ];
```

```
## 取得列, 按顺序取列
```

```
tb3[, c("z", "y")];
```

```
## 替换列
```

```
tb3[["z"]] <- c(4.6, 5.5);
```

```
## 替换行
```

```
tb3[ 1, ] <- tibble( x = "d", y = 20, z = 46, a = 10, b = "C", c = "LILI" );
```

2.8 tibble 与 data.frame 之间相互转换

```
library(tibble)
head( as_tibble(iris) );
```

```
# A tibble: 6 x 5
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
	<dbl>	<dbl>	<dbl>	<dbl>	<fct>
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

note: iris data set gives the measurements in centimeters of the variables sepal length and width and petal length and width, respectively, for 50 flowers from each of 3 species of iris (鸢尾属植物). The species are Iris setosa, versicolor, and virginica.



tibble to dataframe

```
library(tibble)
as.data.frame( head( as_tibble(iris) ) );
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

2.9 differences between tibble and data.frame

tibble evaluates columns sequentially

```
rm(x,y); ## 删除可能存在的 x , y
tibble(x = 1:5, y = x ^ 2); ## 可以用 tibble 这样做
```

```
# A tibble: 5 x 2
```

	x	y
	<int>	<dbl>
1	1	1
2	2	4
3	3	9
4	4	16
5	5	25

练习:

```
data.frame(x = 1:5, y = x ^ 2); ## 但 data.frame 不行
```

```
Error: object 'x' not found
```

differences between tibble and data.frame, cont.

data.frame 在取 **subset** 操作时, 会造成困扰

```
df1 <- data.frame(x = 1:3, y = 3:1);
class(df1[, 1:2]);

[1] "data.frame"

## subset 操作: 取一列, 期待得到一个 data.frame ()
class(df1[, 1]); ## 结果得到一个 vector ...

[1] "integer"

## 而 tibble 则不会
df2 <- tibble(x = 1:3, y = 3:1);
class(df2[, 1]); ## 永远都是 tibble

[1] "tbl_df"      "tbl"        "data.frame"
```

differences between tibble and data.frame, cont.

tibble 可以进行可控的数据类型转换:

```
class(df2[[1]]); ## 取一列, 转换为 vector
```

```
[1] "integer"
```

```
class(df2$x); ## 用 [[ ]] 或 $ 都可以哦
```

```
[1] "integer"
```

differences between tibble and data.frame, cont.

recycling

```
data.frame(a = 1:6, b = LETTERS[1:2]); ## data.frame 可以!!!
```

```
  a b
1 1 A
2 2 B
3 3 A
4 4 B
5 5 A
6 6 B
```

```
tibble(a = 1:6, b = LETTERS[1:2]); ## 但 tibble 不行!!!
```

```
Error in `tibble()` :
! Tibble columns must have compatible sizes.
* Size 6: Existing data.
* Size 2: Column `b`.
i Only values of size one are recycled.
```

注意 tibble 的 recycling 仅限于长度为 1 或等长; 而 data.frame 则为整除即可。

practises for recycling

```
tibble(a = 1, b = 1:3);
```

```
# A tibble: 3 x 2
```

```
      a      b
  <dbl> <int>
1     1     1
2     1     2
3     1     3
```

```
tibble(a = 1:3, b = 1);
```

```
# A tibble: 3 x 2
```

```
      a      b
  <int> <dbl>
1     1     1
2     2     1
3     3     1
```

```
tibble(a = 1:3, c = 1:2);
```

```
Error in `tibble()`:
```

```
! Tibble columns must have compatible sizes.
```

```
* Size 3: Existing data.
```

differences between tibble and data.frame, cont.

data.frame will do partial matching

```
df <- data.frame(abc = 1)
df$ab; ## unwanted result ...
```

```
[1] 1
```

```
## -- but tibble will never do it;
```

```
df2 <- tibble(abc = 1)
```

```
df2$a; ## produce a warning and return NULL
```

```
NULL
```

2.10 data.frame 和 tibble 的高级使用技巧

attach 和 detach

```
head( iris, n = 3 );
```

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa

```
head( iris$Sepal.Length , n = 10 ); ## 用 $ 操作符取得一列 ...
```

```
[1] 5.1 4.9 4.7 4.6 5.0 5.4 4.6 5.0 4.4 4.9
```

```
attach( iris );
```

```
head( Sepal.Length , n = 10 ); ## 直接用列名获取数据;
```

```
[1] 5.1 4.9 4.7 4.6 5.0 5.4 4.6 5.0 4.4 4.9
```

```
detach(iris); ## 取消 attach 操作 --
```



with 函数

```
with( iris, head( Sepal.Length, n = 10 )); ## 用 with 也可以实现
```

```
[1] 5.1 4.9 4.7 4.6 5.0 5.4 4.6 5.0 4.4 4.9
```


within 函数

也可以用 within 对多列数据进行修改

```
head( airquality , n = 3 );
```

	Ozone	Solar.R	Wind	Temp	Month	Day
1	41	190	7.4	67	5	1
2	36	118	8.0	72	5	2
3	12	149	12.6	74	5	3

```
aq <- within(airquality, {      # Notice that multiple vars can be changed
  lOzone <- log(Ozone)
  Month <- factor(month.abb[Month])
  cTemp <- round((Temp - 32) * 5/9, 1) # From Fahrenheit to Celsius
  S.cT <- Solar.R / cTemp # using the newly created variable
  rm(Day, Temp) ## 删除特定列 ...
});
```

```
head(aq, n = 3 );
```

	Ozone	Solar.R	Wind	Month	S.cT	cTemp	lOzone
1	41	190	7.4	May	9.793814	19.4	3.713572
2	36	118	8.0	May	5.315315	22.2	3.583519



Section 3

section 3: file IO: read a file into tibble & write tibble to a file

read from files

使用 functions from the readr package

```
## readr is part of tidyverse  
library(tidyverse); ## or alternatively  
library(readr);
```

available functions

- read_csv(): comma separated (CSV) files
- read_tsv(): tab separated files
- read_delim(): general delimited files
- read_fwf(): fixed width files
- read_table(): tabular files where columns are separated by white-space.
- read_log(): web log files

read a file into tibble

```
myiris <- read_csv("data/talk03/iris.csv");
```

注意 输出的 columns 定义

read with predefined column types

```
myiris2 <- read_csv("data/talk03/iris.csv", col_types = cols(  
  Sepal.Length = col_double(),  
  Sepal.Width = col_double(),  
  Petal.Length = col_double(),  
  Petal.Width = col_double(),  
  Species = col_character()  
));
```

how to read from other formats??

try the following packages for other formats

- **haven** - SPSS, Stata, and SAS files
- **readxl** - excel files (.xls and .xlsx)
- **DBI** - databases
- **jsonlite** - json
- **xml2** - XML
- **httr** - Web APIs
- **rvest** - HTML (Web Scraping)

write to files

use the following functions to write object(s) to external files

- Comma delimited file: **write_csv**(x, path, na = "NA", append = FALSE, col_names = !append)
- File with arbitrary delimiter: **write_delim**(x, path, delim = " ", na = "NA", append = FALSE, col_names = !append)
- CSV for excel: **write_excel_csv**(x, path, na = "NA", append = FALSE, col_names = !append)
- String to file: **write_file**(x, path, append = FALSE)
- String vector to file, one element per line: **write_lines**(x,path, na = "NA", append = FALSE)
- Object to RDS file: **write_rds**(x, path, compress = c("none", "gz", "bz2", "xz"), ...)
- Tab delimited files: **write_tsv**(x, path, na = "NA", append = FALSE, col_names = !append)

练习

```
## write iris to outfiles of various formats  
write_csv( iris, "iris.csv" );  
write_tsv(iris, "iris.tsv", quote_escape = "none");
```

check readr cheatsheet (please Google).

Section 4

section 4: 练习 & 作业

小结

今次提要

- ① data.frame, tibble
- ② 定义、区别、转化
- ③ read files from harddrive (IO)

下次预告

- factor : R 另一个超级重要且难以上手的概念
- 基础和进阶绘图 (配合 factor 讲解)

important

- all codes are available at Github:
<https://github.com/evolgeniusteam/R-for-bioinformatics>

练习 & 作业

- Exercises and homework 目录下 talk03-homework.Rmd 文件;
- 完成时间: 见钉群的要求